

VPN-LAN Service Access — Integration Status

Revision: 3 **Last modified:** 2026-07-01T19:05:00Z **Status:** In progress — ALL 13 phases (0-12) have committed deliverables on main. **Phase 12 bidirectional exposure** (operator mandate) is landed: design (`bidirectional_exposure.md`) + autonomous ingress-allowlist teeth (default-deny + narrow allowlist, `INGRESS_MUT` caught 3/3) + both-way reverse-leg assertions in all 5 protocol tests (NFS NLM/NSM callback, FTP active, Cast callback, adb reverse, email N/A). Autonomously PROVEN + wired into the standing suite as §11.4.135 guards: Phase-1 SSRF carve-out teeth, Phase-12 ingress teeth, and the autonomous battery (stress+chaos 100-iter, benchmark+memory, concurrency+load, DNS-rebinding-gap) — ALL GREEN. Phases 2/3/4/5/6/7/10/11 protocol/reflector/Cast/ADB/containerization/Challenge+HelixQA assets are AUTHORED + honest-SKIP-proven (bridge-down ⇒ SKIP + exit 0, zero fake PASS); every LIVE round-trip (both directions) is operator-gated on the sword connection. control-plane coverage raised (acl-helper 53.8% / cmd/api 61.1% / internal/redis 61.2% / cmd/healthd 70.3%, all -race clean). **Data-plane env-block (operator-actionable, NOT a code defect):** host rootless-podman aardvark-dns cannot bind the netavark gateway :53 → :53128/:51080 don't serve + ./start fails; the standing suite now honestly reports this (3 env FAILs) instead of hanging. Every claim below cites a real committed file/qa-results/ artefact (§11.4.6); live paths honestly SKIP until the operator connects the bridge (§11.4.3) + repairs podman. **Authority:** Inherits constitution/Constitution.md per §11.4.35. §11.4.45 integration-status doc for the VPN-LAN service-access feature workstream (feature/vpn-aware-dynamic-routing). **Companion:** summary [Status_Summary.md](#) · plan [PLAN.md](#) · Miracast verdict [miracast_verdict.md](#) · operator guide [../..guides/vpn_lan_bridge_setup.md](#)

Operator-blocked / pending items (read first — §11.4.45 O(1) surface)

Item	Why blocked / pending	Unblock condition
Live protocol	Need the live sword bridge connection —	Operator copies <code>.env.example</code> → <code>.env</code> ,

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round-trips (SMB/NFS/FTP/SFTP/WebDAV/email/Cast/ADB)	secrets + Mullvad WG + root/sudo on the bridge side (§11.4.21 / §11.4.10). The autonomous slate proves the honest-SKIP path + the anti-bluff gate now; the live round-trip evidence exists only when the bridge is genuinely up.	points it at sword_toolkit, brings the VPN up; then each test emits real captured round-trip evidence.
Phase 5 discovery reflector — remote deploy	Deploying an Avahi/SSDP reflector changes a remote host (§11.4.122) — requires an interactive keep/deploy decision. Local reflector config + a local-stub discovery test are autonomous.	Operator authorizes the remote-side reflector deployment.
Phase 7 ADB flash (usbip fastboot) on real hardware	Flashing a real device is high-blast-radius (§11.4.133) — operator-gated. adb connect/shell/getprop over routed 5555 is autonomous once the bridge is up.	Operator authorizes the specific device flash.

Phase status matrix (captured-evidence-driven — §11.4.5/§11.4.69)

Phase	Scope	Status	Evidence / commit
0	env-var sword bridge scaffold + sword-doctor preflight	PASS	d781002; sword-doctor 3-verdict discrimination (UP/SKIP/MISCONFIGURED) + standing-suite test_vpn_lan_bridge GREEN (honest SKIP + §11.4.115 teeth PASS, qa-results/suite/run_20260701T160156Z.log 71/64/7/0) 5c28f56
1	L3 routed gateway + SSRF allowlist reconciliation	AUTHORED (local-stub GREEN)	1d87b3a tests/vpn_lan/ssrf_carveout_teeth.sh: proves the SOCKS SSRF floor survives the narrow 10/8

Phase	Scope	Status	Evidence / commit
			carve-out with NO live VPN + sockd.conf READ-ONLY — T1 live-floor teeth (metadata+loopback+all RFC1918 blocked, public passes), T2 narrow-carve floor-preserving, T3 ordering (first-match-wins); SSRF_MUT=1 catches 2/2 golden-bad collapses (§11.4.107(10)); wired into standing suite test_vpn_lan_ssrf. Live 10/8 carve-out application stays bridge-gated + operator-gated (§11.4.101/§11.4.133)
2	SMB/CIFS/NMB + NFS round-trip	AUTHORED (SKIP-proven)	182e80a tests/vpn_lan/smb_nfs_roundtrip.sh; bridge-down SKIP + exit 0, PASS_lines=0 (no fake PASS); live sha256 round-trip operator-gated
3	FTP/FTPS/SFTP + WebDAV (WebDAV via existing Squid)	AUTHORED (SKIP-proven)	182e80a tests/vpn_lan/ftp_sftp_webdav.sh; wrong-answer⇒FAIL (WebDAV non-207 / SFTP sha-mismatch), unreachable⇒SKIP; live operator-gated
4	IMAP/IMAPS/SMTP-submission/POP3S + open-relay guard	AUTHORED (SKIP-proven)	2f31460 tests/vpn_lan/email_roundtrip.sh; open-relay negative test (external-RCPT-accepted⇒FAIL); creds via stdin never argv (§11.4.10); live operator-gated
5	mDNS/SSDP/WS-Discovery/DNS-SD reflector	AUTHORED (SKIP-proven)	d0b42df reflector_design.md (Avahi enable-reflector + SSDP LOCATION-rewrite, cited RFC 6762/6763/5771/2365) + tests/vpn_lan/discovery_reflect.sh (bridge-down SKIP + exit 0, PASS_lines=0); remote deploy operator-gated (§11.4.122)
6	Chromecast / DIAL (reflect discovery + route control)	AUTHORED (SKIP-proven)	65043ce tests/vpn_lan/chromecast_dial.sh; eureka_info:8008 (200+JSON name⇒PASS, 200-no-name/non-200⇒FAIL fail-closed),

Phase	Scope	Status	Evidence / commit
7	ADB over VPN (access/debug/connect/flash)	AUTHORED (SKIP-proven)	CASTV2 :8009 liveness via status transition (identical⇒SKIP not fake-PASS §11.4.107); discovery = Phase-5 dependency; live operator-gated 65043ce tests/vpn_lan/adb_over_vpn.sh; routed 5555 connect+getprop⇒PASS, offline/unauthorized⇒FAIL; §11.4.174 device-safety (disconnect-our-serial-only, never kill-server); flash via usbip USB-bound operator-gated (§11.4.133); in-depth phased design in flight
8	Miracast structural-impossibility verdict	PASS (Won't-fix)	12faf12 miracast_verdict.md (§11.4.112); cited Wi-Fi-Alliance evidence; Google Cast as the routable alternative; no fake traversal test
9	Documentation (operator bridge-setup guide)	PASS	a5e5616 ../../guides/vpn_lan_bridge_setup.md (routing map + verdict table + 12-row protocol matrix + FAQ); HTML+PDF §11.4.168 leak-clean
10	Containerization via containers submodule (§11.4.76)	AUTHORED (SKIP-proven)	1911d5e containerization.md + vpn_lan_containers.yaml (project-side service decl, config-injected §11.4.28) + tests/vpn_lan/container_boot.sh (bridge-down SKIP exit 0; malformed decl⇒FAIL fail-closed; never boots a container); submodules/containers untouched; remote deploy operator-gated (§11.4.122)
11	§11.4.169 full test-type coverage + Challenges + HelixQA	AUTHORED (mostly GREEN)	89f73b7 challenges/scripts/run_vpn_lan_challenges.sh + HelixQA tools/helixqa/banks/vpn_lan.yaml (9 cases); autonomous test types LANDED + GREEN — stress+chaos (97d9733, 100-iter identical hash), benchmark+memory + DNS-rebinding gap (4e2810c),

Phase	Scope	Status	Evidence / commit
12	Bidirectional exposure + ingress allowlist (operator mandate)	AUTHORED (GREEN + suite-wired)	concurrency+load (7e73b08); HelixQA-run honestly blocked (6 un-vendored siblings + podman, §11.4.3) 2ed0fed bidirectional_exposure.md (return-route model + per- protocol both-way table + inverted-posture ingress = mirror of egress floor) + tests/ vpn_lan/ ingress_allowlist_teeth.sh (default-deny/exact-permit/ host+port-narrow, INGRESS_MUT caught 3/3, PASS=4); cc8b620 both-way reverse-leg assertions in all 5 protocol tests (NFS NLM/NSM, FTP active, Cast callback, adb reverse, email N/A §11.4.6); live both-way round-trip operator-gated
—	§11.4.135 standing regression guards (autonomous, no bridge/ podman)	GREEN	6d8893f test_vpn_lan_ssrf + test_vpn_lan_ingress + test_vpn_lan_autonomous_battery wired into run-tests.sh; all rc=0 PASS≥1 FAIL=0 in the standing suite

Honest boundary (§11.4.6)

The feature's design + decoupled bridge + anti-bluff test scaffolding are proven now: the Phase-0 bridge is GREEN in the standing suite with a genuine up↔down discrimination (§11.4.115 teeth), and Phases 2/3/4 tests refuse to fake a PASS when the bridge is down (bridge-down ⇒ honest SKIP, exit 0, zero ^PASS: lines) — the anti-bluff gate is the value delivered autonomously. Miracast is honestly classified structurally-impossible (§11.4.112) with Google Cast as the routable alternative. The **live** protocol round-trip evidence (real bytes, sha256, mailbox content, device serials) exists only once the operator connects the sword bridge (secrets + Mullvad + root/sudo) — until then those paths honestly SKIP (§11.4.3), never a fake PASS. No phase is "done" for the live capability until its runtime signature verifies with captured evidence on a genuinely-up bridge (§11.4.108) and it crosses the

§11.4.169 test-type matrix. This doc is §11.4.45-synced + §11.4.65-exported; it does not substitute for the §11.4.40 full-suite retest before any release tag.